

Principles of Programming Languages Compilation

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Outline

Introduction

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Can theorem provers help?

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Demo(s)

- Expressions

- Defining a stack machine

- Compilation & Soundness

What are compilers?

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- ▶ *source* code $\rightarrow$ *target* code
- ▶ Compilers facilitate the work of the programmer: programmers focus only on the task to be solved rather than dealing with cumbersome low-level programming

Compilers short history

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- ▶ 1957, IBM: the first complete compiler of FORTRAN
- ▶ Later, COBOL was compiled on multiple architectures
- ▶ 1962, MIT: compiler for Lisp
- ▶ 1970s: compilers for a languages started to be implemented in the same language

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- ▶ code generation

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- ▶ Challenge: make sure that compilers generate programs that are equivalent with the intended behaviour of the compiled programs
- ▶ Compilers can have bugs!
 - ▶ <https://web.cs.ucdavis.edu/~su/publications/issta16-compiler-bug-study.pdf>
 - ▶ Year 2016, found 39890 bugs in GCC, 12842 in LLVM
 - ▶ Most buggy code: C++ component

Compilers bugs

- ▶ The expression $(x == c1) \vee (x < c2)$ is equivalent to $x < c2$ if $c1$ and $c2$ are constants and $c1 \leq c2$

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- ▶ LLVM bug: $(x == 0) \vee (x < -3)$ was transformed into $(x < -3)$ due to an unsigned comparison

Source: <https://www.cs.cornell.edu/courses/cs6120/2019fa/blog/bug-finding/>

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 - ▶ Make sure proofs are **exact** - computer checked

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- ▶ These can be used to define compilers too and then prove properties about them!

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Expressions

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Simple expressions in Coq

- ▶ Simple language of expressions - BNF:

$$\text{Exp} ::= \text{Nat} \mid \text{Id} \mid \text{Exp} \text{ "+" } \text{Exp} \mid \text{Exp} \text{ "*" } \text{Exp}$$

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- ▶ (Abstract) syntax can be defined using `Inductive`:

```
Inductive Exp :=  
  | const : nat -> Exp  
  | id : string -> Exp  
  | plus : Exp -> Exp -> Exp  
  | times : Exp -> Exp -> Exp.
```

Interpreter

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- ▶ The interpretation function:

```
Fixpoint interpret (e : Exp)
    (sigma : Var -> nat) : nat :=
match e with
| const x => x
| id s => sigma s
| plus e1 e2=>(interpret e1 sigma)+(interpret e2 sigma)
| times e1 e2=>(interpret e1 sigma)*(interpret e2 sigma)
end.
```

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Exercise: a stack machine

► Syntax:

```
Inductive Instruction :=  
| push_const : nat -> Instruction  
| push_var   : Var -> Instruction  
| add        : Instruction  
| mul        : Instruction.
```

Exercise: a stack machine

► Semantics:

```
Fixpoint run_instruction (i : Instruction)
  (env : Var -> nat) (stack : list nat) :=
match i with
| push_const n => n :: stack
| push_var v  => (env v) :: stack
| add         => match stack with
  | n1 :: n2 :: s' => (n1 + n2) :: s'
  | _              => stack
  end
| mul         => match stack with
  | n1 :: n2 :: s' => (n1 * n2) :: s'
  | _              => stack
  end
end.
```

Exercise: a stack machine

► Run instructions:

```
Fixpoint run (instr : list Instruction)
  (env : Var -> nat)
  (stack : list nat) : list nat :=
match instr with
| nil => stack
| i :: is' => run is' env
               (run_instruction i env stack)
end.
```

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Compile

► Compilation of `Exp` to the stack machine:

```
Fixpoint compile (e:Exp) : list Instruction :=  
  match e with  
    | const n => push_const n :: nil  
    | id v => push_var v :: nil  
    | plus e1 e2 => (compile e1) ++ (compile e2)  
                   ++ (add :: nil)  
    | times e1 e2 => (compile e1) ++ (compile e2)  
                   ++ (mul :: nil)  
  end.
```

Compilation results

► Compilation correct:

Theorem `compilation_correct`:
 forall e env,
 run (compile e) env nil =
 (interpret e env) :: nil.

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forall e env,  
  run (compile e) env nil =  
    (interpret e env) :: nil.
```

► Invariant:

Lemma `compilation_correct'`:

```
forall e env is' stack,  
  run (compile e ++ is') env stack =  
    run is' env (interpret e env :: stack).
```

Bottom line

- ▶ We have defined two languages in Coq: a language of expressions Exp and a stack machine
- ▶ We show how Exp can be compiled to the stack machine
- ▶ We **prove** compilation correct!

CompCert - I

- ▶ X. Leroy, S. Blazy, Z. Dargaye, J.B. Tristan; et al.
- ▶ Link: <http://compcert.inria.fr/>
- ▶ Project: develop and prove a realistic compiler, usable for critical embedded software
- ▶ Source language: a very large subset of C
- ▶ Target language: PowerPC/ARM/x86 assembly
- ▶ Generates compact and fast code; careful code generation + some optimizations
- ▶ Compiler written from scratch + **proof**

CompCert - II

- ▶ 50000 lines of Coq
- ▶ 4 person-years
- ▶ Trusted execution, certified compilation, static analysis (Verasco)
- ▶ Direct and indirect project partners: Airbus, AbsInt, etc.
- ▶ Airbus: flight control software of the A380
- ▶ MTU: critical control system in diesel engines that are deployed in nuclear power plants
- ▶ Institute of Flight System Dynamics at TUM: development of flight control and navigation algorithms

Resources

1. Chapter Small-Step Operational Semantics in **Software Foundations - Volume 2**, Benjamin C. Pierce, Arthur Azevedo de Amorim, Chris Casinghino, Marco Gaboardi, Michael Greenberg, Cătălin Hrițcu, Vilhelm Sjöberg, Andrew Tolmach, Brent Yorgey

Section “*A Small-Step Stack Machine*”:

<https://softwarefoundations.cis.upenn.edu/plf-current/Smallstep.html#lab169>